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TEAM NAME : SimpleX Crew

COLLEGE NAME : Sri Venkateswara College of Engineering

PROBLEM STATEMENT : Digital Booking Platform for Farm Machinery and Agri Vehicles

PROBLEM STATEMENT

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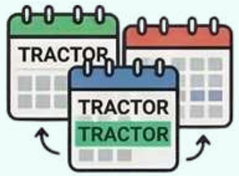
1 CURRENT PROBLEMS IN RURAL FARMING

Few owners (tractors, harvestors)

Most farmers depend on renting

Manual booking (phone / visits)

Time - consuming to check availability



Booking conflicts can occur

Lack of transparency (pricing, damage)

2 IMPACT ON FARMERS AND OWNERS

Loss of valuable farming time

Reduced Crop Productivity

Underutilization of Machinery

Payments & Damage disputes

Lack of Trust

3 NEED FOR A DIGITAL PLATFORM

Easy search for nearby machinery

Real-time visibility of availability

Online Booking & Scheduling

Secure Damage & Protection

Vehicle tracking & Transparency

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SOLUTION

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DIGITAL BOOKING PLATFORM FOR FARM MACHINERY AND AGRI VEHICLES

- Notification and Alert System
- Real-Time Equipment Availability & Smart Booking System
- Flexible Rental Options (Hourly & Daily Booking)
- GPS-Based Equipment Discovery & Live Vehicle Tracking
- Secure Payment System (Pre-Payment & Partial Payment)
- Damage Protection Policy & User Accountability
- User-Friendly Multi-Language Mobile Application
- Admin Dashboard & Platform Management
- Offline/Low Network Booking Mode

Existing Solution	VS	*Proposed System*
Manual booking process		Digital booking platform
No availability tracking		Real-time availability tracking
No Damage Policy		Structured damage protection
Unclear pricing system		Secure payment system

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FLOW DIAGRAM

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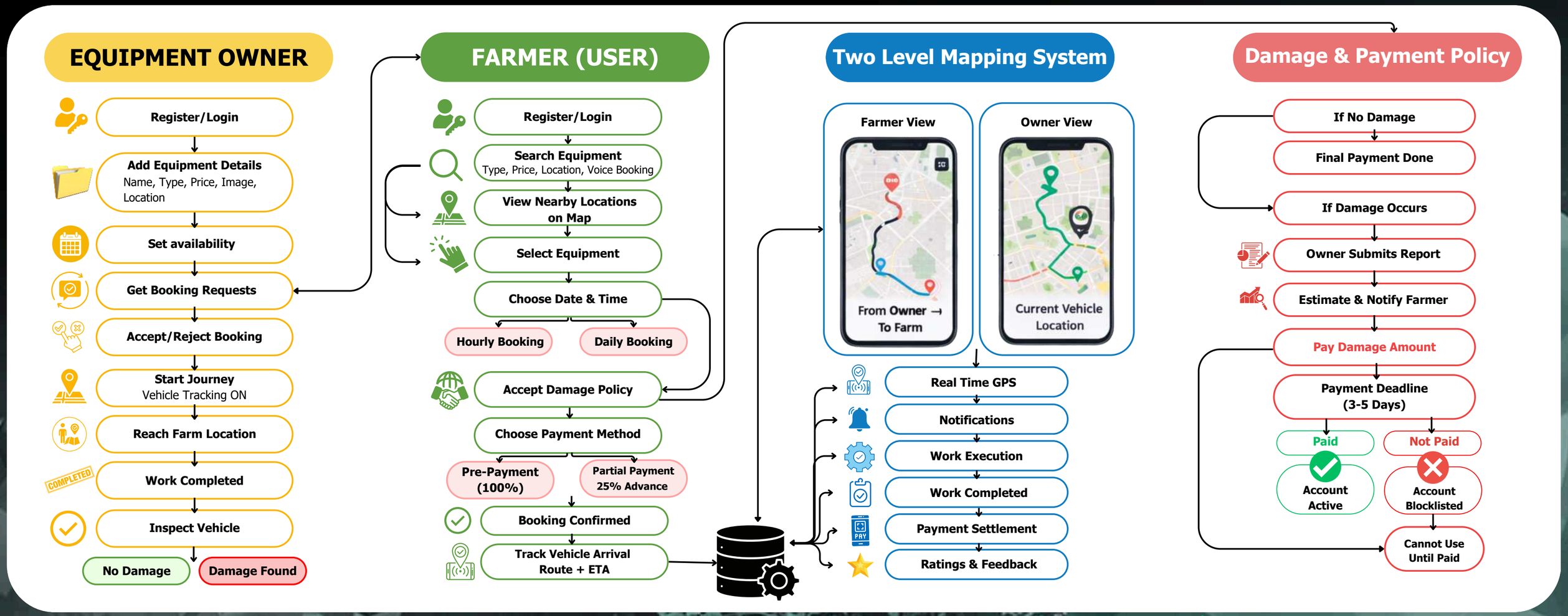

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TECHNICAL REQUIREMENTS

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1.USER INPUTS

Farmer App

Voice Commands (🎙) & Touch Input

Owner App

Fleet Dashboard & Equipment Listing

2.GATEWAY & AUTHENTICATION

API Gateway

JWT

JWT Authentication, FCM Messaging, SQLite Offline Sync

3.ORCHESRATION LAYER (THE BRAIN)

Voice NLP Agent

Process Text → Extract Intent

Booking State Machine

Manage Lifecycle, Hourly/Daily Logic, Group Booking

Geospatial Engine

WebSocket Sessions, Radius Matching

4. INTELLIGENCE & DATA

AI Forecast & RECs Engine

Demand Prediction & Equipment suggestion

PostgreSQL + PostGIS

Spatial Data, Transactions, User Data

Redis Live Cache

High Frequency Live Location Pub/Sub

5.OUTPUT LAYER

Escrow Payment Engine

Handles Advances, Penalties, Final Statements

Live Map Tracker

Real-Time Vehicle Tracking (Farmer/Owner)

FCM Alert Dispatcher

Booking Confirmation & Arrival Alerts

EXTERNAL INTEGRATIONS (APIs)

- Google Maps API (Rounting)
- Weather API (Forecast)
- Payment API (Razorpay)

FRONTEND

Flutter | Tailwind CSS | React.js

Flutter · Tailwind CSS · React.js

DATA & GEOSPATIAL

PostgreSQL | Razorpay | Google Maps API

PostgreSQL · Razorpay · Google Maps API

BACKEND & LOGIC

Node.js | WebSockets | Python

Node.js · WebSockets · Python

CLOUD & SECURITY

JWT | Firebase (FCM) | AWS

JWT · Firebase · AWS

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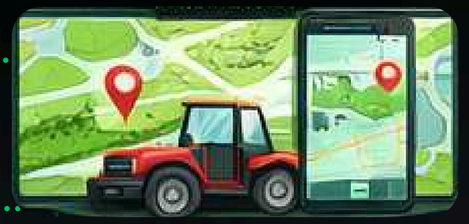
NOVELTY / USE CASE

NOVELTY

- *Digital Platform for Rural Equipment Sharing*

- *Real-Time Tractor Availability*

- *Two-Level GPS Mapping System*



- *Flexible Booking Options*

- *Smart Damage Protection policy*

- *Secure Dual Payment System*

USE CASE

- *Beneficial for Small Farmers*

- *Increased Revenue for Equipment Owners*

- *Village-Level Equipment Sharing*

- *Emergency Farming Needs*

- *Transparent Farming Operations*

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BUSINESS MODEL / TARGET MARKET

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TARGET AUDIENCE

FARMERS

- Easy Tractor Access
- Affordable Machinery Rental
- Transparent pricing
- Real time booking

Equipment owners

- Extra income
- Better machine utilitzation
- Organized booking
- Digital Payment Tracking

BUSINESS MODEL

- Commission on every booking
- Owners can upgrade to premium plans.
- equipment Promotion
- Service fee

Premium features:

- Higher visibility in search
- Priority booking requests
- Analytics dashboard

Value Creation

For Farmers

- Quick machinery access
- Fair pricing
- Less waiting time

For Owners

- More customers
- Higher machine usage
- Secure payments

For Platform

- Revenue through bookings
- Scalable digital agriculture ecosystem

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